PROJECT CLINIC

Some Pitfalls to Pay Attention to

Reality Checks: Early and Often

- A (fun!) project about offensive words in movies over time
  - Used a list of offensive words off the internet
  - Lots and lots of analysis, some surprising results

Reality Checks: Early and Often

–At the very end:

Look at your data!!!

Reality Checks: Early and Often (2)

- A project about Airbnb and nearby tourist attractions
  - “One map that left us a bit surprised was the map of Rio de Janeiro, since the assets were all far from the attraction and the time it takes to get there by transit is more than an hour.”

- They searched “Christ the Redeemer”
- From a quick search, most entries are in Portugese and use "Cristo Redentor" instead

Ignoring Reality

- An app that recommends changes to bus companies to minimize delays
  - Step 1: build a model to predict delay
  - Step 2: Given a specific bus ride, suggest changes
  - (Step 3: Profit)

- Possible recommendations: change the month, change the temperature

Ignoring Reality (2)

- Project description: predict the results of a sports match (yay!)

- Actual project: give the classifier data from the entire match

Omitting Critical Details

- Prediction task between 10 labels, but no way for me to understand what they are without looking at the code. :(
- “We merged data from various recent sources”
- “we got better accuracy with [technique]”
- “huge amount of reviews”
- “We run community detection”

Magic Numbers

- “We set cutoff to 40 minutes”…
- “We had a train/val/test split of 64/16/20”
- "We define success if we could predict within a range of 50 points from the curve."

- In general, either **justify** your choices or show how **stable** the algorithm is for different choices (or both)

Don’t Just Report Accuracy

- Look at the model mistakes
  - Can you say something interesting?
- Confusion matrix might help

Super Vague Evaluations

- The sentence "We let some people try the system. Overall, feedback was positive" is **not** an evaluation
  - Also "there was an increase in the amount of likes in each 10 iterations" – from what to what?

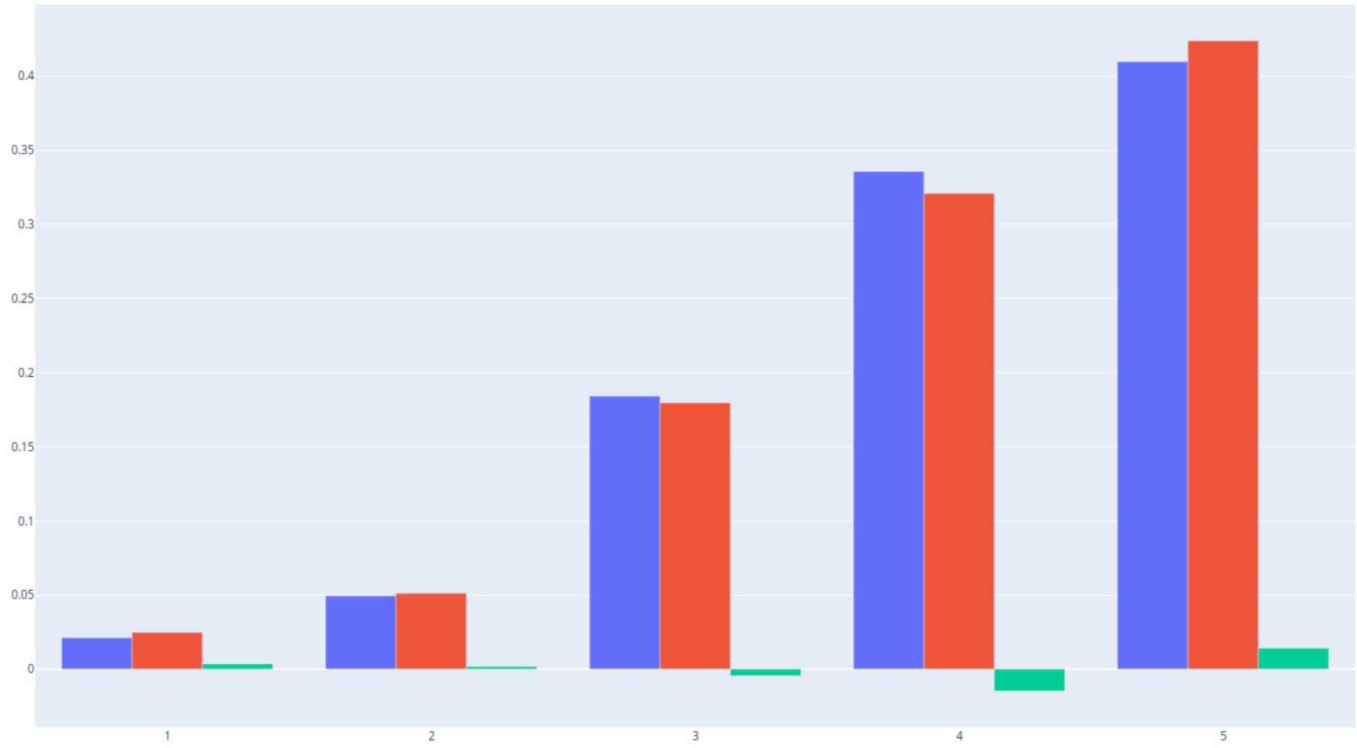
- Try getting some **numbers**
  - **Compared** to baselines!
- **Quotes** are also helpful (think out loud)
  - What's good? What's bad? What did you notice?

Forgetting the Takeaway Message

- “Here is a viz”
- Write what I’m supposed to learn from looking at it

Don’t Overstate the Results

- “These results clearly show that group A is better!”



Green = difference

Don’t Cherrypick

- “We downloaded 15000 movie scripts”
- … and ended up analyzing one

- In-depth analysis is fine, but if you have an algorithm try it on several (very different!) items
  - Can you say something about when it works better?

Limitations and Bias

- Really think about limitations of the data and how they could affect the results
  - E.g., if there are tags, who gets to choose them?

Model choice

- State parameters
- And justify

Visualizations

- In a legible font!
- Axes, titles
- Discuss outliers

Effort can come from multiple places

- E.g., Kaggle vs. super hard crawling

Have Fun!

- Remember, the idea is to play with data and get something interesting out :)